

# Density-gated noise rectification in a Vicsek–Couzin flock: one dense-phase mechanism, additive at matched decorrelation

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## Abstract

Flocking models hold the self-propulsion speed and the angular-noise distribution fixed as global constants. Active matter has relaxed each of them on its own: a density-dependent speed drives motility-induced phase separation, and heavy-tailed Lévy reorientations are reported in bacteria, in foragers and in migrating cells. What has not been tested is whether the two reinforce each other when one cue drives both, and the question is as much methodological as physical, since apparent cooperation between adaptive channels is easy to manufacture and hard to rule out. We therefore gate the speed and the stability index of a wrapped  $\alpha$ -stable kick through a single sigmoid in the local neighbour count, so that a crowded particle is at once slow and Gaussian-like and an isolated one fast and Cauchy-like: the arrangement most favourable to a synergy. None survives. Heavy-tailed reorientations alone grow no size-dependent susceptibility; only the motility channel builds a phase-separated dense phase, and the noise-shape channel acts solely inside it, rectifying wrapped Cauchy kicks into Gaussian ones. The apparent super-additivity comes from comparing modes at matched nominal noise amplitude instead of matched effective decorrelation: equalise the effective rate and the susceptibility excess is gone, while the dense static and dynamic correlation gaps reverse sign, leaving two channels that simply add. The coupling thus reduces to a density-gated lowering of the dense-phase effective noise, set by the motility threshold alone and contingent on a metric kernel. The lesson travels beyond this model: heavy-tailed and Gaussian dynamics have to be compared at matched decorrelation, not matched amplitude, or additivity will be read as synergy.

**Keywords:** active matter, self-propelled particles, classical phase transitions, finite-size scaling

## 1 Introduction

The Vicsek model [1] freezes two ingredients that real collectives keep fluid: every particle moves at the same speed, and every reorientation is drawn from the same Gaussian. Neither assumption survives contact with data. The sign of the departure, on the other hand, varies from system to system, and that variability is part of the evidence rather than a detail to be smoothed over. Motility does respond to crowding, and it responds downward in confluent tissue and in traffic: epithelial monolayers slow as they approach jamming [2, 3], and vehicular flow carries a pronounced speed–density anti-correlation [4, 5]. In bacterial suspensions and swarms the sign reverses, hydrodynamic and steric coupling raising the mean swimming speed as density grows

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[6, 7]; even in monolayers the slowdown has been traced to the maturation of cell-cell contacts rather than to density as such [8]. The turning statistics are no closer to a fixed Gaussian. Heavy-tailed Lévy reorientations are documented in foragers, T cells and swarming bacteria [9–12], and crowding reshapes those too: cell density suppresses the reversal frequency of *Myxococcus xanthus* through contact-dependent signalling [13], raises the effective reorientation rate of *Escherichia coli* in dense suspensions [6], and shifts the Lévy exponent of swarming trajectories as a swarm densifies [7]. Even there the interpretation is contested, the best-studied case of a density-driven Lévy-to-Brownian shift having been attributed to passive encounter statistics rather than to any change of movement strategy [14]. A flocking model in which both inputs read the local environment is therefore closer to the biology than one that fixes either, even if no single organism is known to realise the particular combination we study.

These two relaxations have so far been pursued apart. On the motility side, a steep enough density-dependent speed drives motility-induced phase separation (MIPS) with no alignment at all [15–20], now a standard route to clustering in active matter, and the same slowdown underlies motility-driven jamming in tissue [3]. That channel has already been combined with alignment, which bears directly on what follows: Farrell et al. [21] couple a density-dependent motility to polar alignment and obtain bands and asters, while Solon et al. [22] and Solon et al. [23] show that the banded state of flocking models is a microphase separation sustained by nonequilibrium fluctuations. Our motility-only ablation belongs to that established line and we treat it as a reference point rather than as a finding. On the noise side, the Vicsek lineage has established that how the noise is implemented, and not merely how large it is, governs the transition: angular and vectorial noise behave differently and carry very different crossover sizes [24, 25], the apparent order of the transition turns on that choice [26–29], and the measured exponents are not universal but drift with speed, density and interaction radius [30–33]. Alignment has likewise been combined with active, non-thermal fluctuations [34, 35] and recast in spin language to sharpen the transition question [36], while heavy-tailed angular noise has run on its own Lévy-driven track [11, 37, 38]. Across all of it, and in the reviews that organise the field [39–42], the self-propulsion speed and the noise statistics remain independent modelling choices, each varied while the other is held fixed.

Three things are consequently missing, and they compound. The first is empirical: no collective has been shown to co-regulate its motility and the *tail shape* of its turning noise through a single cue. Density is known to co-regulate speed and reorientation rate [6], to shift heavy-tail exponents [7] and to set reversal frequency [13], but joint regulation of speed and tail shape is an extrapolation from those observations rather than one of them, which is why we treat the shared sensor below as a deliberate limiting case and not as a documented mechanism.

The second gap, on the modelling side, is narrower than it may appear and worth stating precisely, because a density-dependent noise *amplitude* is not new. The vectorial-noise update of Grégoire and Chaté [43] has been implicitly neighbour-count dependent since its introduction; Frouvelle [44] makes both the alignment rate and the noise intensity functions of the local density at the kinetic level; Haghsheini and Mehrafarin [45] scales the angular-noise strength directly by the neighbour count; and Chen and Ma [46] lets local density switch a discrete behavioural mode. What none of these varies is the *shape* of the turning law. Clusella and Pastor-Satorras [47] come closest, generalising Vicsek to an arbitrary angular-noise distribution that may in principle depend on the neighbour count, and then set that dependence aside explicitly as beyond their scope; their distributions are thin-tailed and their speed constant. Our contribution occupies precisely that declined cell. Making the tail index of the turning law a function of local density, and coupling it to a density-dependent speed, lets us ask directly whether the two channels reinforce, cancel or simply add. The question has content, because the two channels meet inside the dense regions a flock builds: the noise distribution sets how a particle escapes a local structure, and the density-dependent speed builds that structure in the first place.

The third gap weighs most on what follows, and it is one of protocol rather than of ingredients. Noise shapes in this literature are compared at matched nominal amplitude, whereas a wrapped  $\alpha$ -stable kick of scale  $\eta$  decorrelates a heading at the rate  $\Gamma = 1 - \exp(-\eta^\alpha)$  [48–50], a quantity

that depends on  $\alpha$  as strongly as it does on  $\eta$ . Varying  $\alpha$  at fixed  $\eta$  therefore varies the effective noise along with its shape, so any cooperation inferred from such a comparison is confounded before it is measured. Closing the first two gaps without controlling for the third would only reproduce that confound at greater computational expense, and it is the control, as much as the coupling, that this paper is about.

We couple the two adaptations in the most economical way available: a single sigmoid in the local neighbour count gates both the self-propulsion speed and the stability index of a wrapped  $\alpha$ -stable angular kick, so a crowded particle is at once slow and Gaussian-like and an isolated one fast and Cauchy-like. The stability index sets how heavy-tailed the turning is, the Cauchy limit ( $\alpha = 1$ ) sprinkling in occasional near-reversals while the Gaussian limit ( $\alpha = 2$ ) keeps every turn small, and we treat the  $\alpha = 1 \rightarrow 2$  interpolation as a family bracketing those two limits rather than as a fit to any organism. Locking both channels to one cue is deliberate. It is the arrangement most favourable to a synergy: if cooperation between an adaptive speed and an adaptive noise shape exists anywhere in this family, it has its best chance of appearing here, which is what makes a negative answer informative. There is no attraction zone, so spatial cohesion comes entirely from the density-dependent motility rather than from a cohesive term, and turning both channels off recovers the original Vicsek dynamics. Two single-channel ablations, a motility-only mode with adaptive speed and fixed Cauchy noise and a noise-shape-only mode with fixed speed and Cauchy-to-Gaussian noise, isolate each contribution against a shared heavy-tailed reference on a single code path, and a nine-size finite-size scan at fixed density ( $\rho_0 = 2.22$ ,  $L = 15$  to 256, ten seeds per size) places the four heavy-tailed modes and the Gaussian Vicsek baseline on the same footing, with the qualitative split reproduced away from that density at  $\rho_0 = 1.0$  and 3.0.

The two channels separate cleanly, and only those carrying the density-dependent speed respond to system size at all. The Cauchy reference and the noise-shape-only mode return slopes statistically indistinguishable from zero ( $a = -0.03$ ,  $[-0.07, +0.01]$  and  $a = -0.02$ ,  $[-0.05, +0.01]$ ), so heavy-tailed reorientations on their own grow no size-dependent susceptibility, their rare bursts washed out among the many neighbours of a dense patch; the motility-only ablation does grow one ( $a = +1.04$ ,  $[+0.97, +1.22]$ ) and the full mode a steeper one ( $a = +1.49$ ,  $[+1.43, +1.68]$ ). That full-mode value calls for a comparison with the literature, not a bare report:  $+1.49$  is statistically indistinguishable from the  $\gamma/\nu = 1.45(2)$  established for the standard Vicsek model [31, 32] on the same  $\chi = N \text{Var}(\varphi)$  convention, whereas the motility-only slope excludes it. We read that agreement as a consistency check and not as evidence of criticality, since the accompanying diagnostics point the other way: the susceptibility peak  $\eta^*(L)$  drifts toward the small- $\eta$  boundary instead of converging on a fixed  $\eta_c$ , no data collapse can be constructed, the Binder cumulant never crosses, and the window  $L = 15$  to 256 straddles rather than clears the crossover size  $L^* \simeq 128$  reported for angular noise at this density [24]. We therefore report these slopes as effective growth rates of a phase-separation susceptibility, bounded above by the  $d = 2$  ceiling  $\chi_{\max} \sim N$  that a single dense domain would saturate, rather than as asymptotic exponents.

Compared at each mode's own near-critical noise, the noise-shape adaptation then appears to lend the motility-driven growth a super-additive boost, an excess  $\Delta_9 = +0.45$  with a 95% seed-bootstrap interval  $[+0.31, +0.60]$ . That excess does not survive the control the effective-noise argument demands. Retuning the motility-only mode so that its heavy-tailed kick decorrelates as fast as the full mode's rectified one drives the excess to  $\Delta_9 = +0.02$ , additive within error, and reverses the dense static and dynamic correlation gaps in the same stroke, the dense  $g(r)$  gap going from  $+0.153$  to  $-0.07$  and the  $C(\tau)$  gap from  $+0.13 \dots +0.24$  to  $-0.08 \dots -0.42$  as the global retune over-quiets the dilute phase. What looked like cooperation was a difference in effective noise, so the two channels act through one local route and, at matched decorrelation, simply add.

What the noise-shape channel does instead is local, and it can be counted. Inside the dense phase the shared sigmoid pushes  $\alpha$  toward 2, and because the kick enters the heading modulo  $2\pi$  the heavy tail acts through the fraction of kicks that overshoot a half-turn and wrap,  $P(|\xi| > \pi) \simeq 0.20\eta$  in the Cauchy limit, which is 1.0% to 6.1% across  $\eta = 0.05$  to 0.3 against less than  $10^{-11}$

for the Gaussian limit at the same scale. Driving  $\alpha$  from 1 to 2 removes precisely this wrapped-uniform fraction, so the Cauchy bursts that would otherwise scramble a slow particle’s alignment are rectified into coherent kicks: the dense-quartile heading correlation gains  $\Delta g = +0.153$  over the motility-only ablation while the dilute quartile shows no gap, the dense mass fuses into fewer and larger clusters, and  $s_{\text{sep}}$  rises from  $\simeq 1.2$  for the fixed-speed references to  $\simeq 1.7$ . A decoupled-sigmoid analysis (Sec. 3.3.4) then pins the control variable, the motility threshold alone setting the sign and amplitude of the rectification ( $R^2 = 0.88$  for bands of constant  $n_\star^v$  against 0.28 for a sequential-timing alternative) with the noise-shape threshold passive, and identifies the metric kernel as a precondition, that dominance weakening to  $R^2 = 0.68$  under topological alignment [51, 52]. The density-triggered slowdown is thus the precondition throughout: without the dense phase it builds, the heavy-tailed rectification has nothing to act on and erodes coherence instead of creating it. We report the internal falsifications alongside, since they bound the claim as much as the positive results do: hysteresis is negligible, the band-soliton index stays at  $b \leq 1.06$  so no travelling band ever forms, and the dense patches consolidate spatially without living any longer (1.60 against 1.57 snapshots,  $p = 0.08$ ).

Section 2 defines the two-feedback model, its ablations and the numerical protocol, and states the matched-decorrelation control on which the rest of the paper leans. Section 3 reports the global diagnostics, the parametric analyses and the local diagnostics (heading correlations, cluster statistics, single-particle dynamics and the decoupled-sigmoid dissection) that localise the rectification to the dense phase. Section 4 discusses the mechanism, its place in the literature and its limits, and the appendix (Sec. A) sketches a heuristic hydrodynamic account of the dense-phase rectification.

## 2 Model and methods

### 2.1 Dynamics

$N$  point particles live in a square box of side  $L$  with periodic boundaries; particle  $i$  carries a position  $\mathbf{x}_i \in [0, L]^2$  and a heading  $\theta_i \in (-\pi, \pi]$ , advanced synchronously through

$$\theta_i(t+1) = [\theta_i^\star(t+1) + \xi_i(t)] \bmod 2\pi, \quad \mathbf{x}_i(t+1) = \mathbf{x}_i(t) + v_i(t+1) \vec{e}_i(t+1) \bmod L, \quad (1)$$

with  $\vec{e}_i = (\cos \theta_i, \sin \theta_i)$ . The desired heading  $\theta_i^\star$  follows the Couzin priority rule [53] with the repulsion set  $\mathcal{R}_i = \{j \neq i : |\mathbf{r}_{ij}| < R_r\}$ ,  $\mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{x}_i$ , and the alignment set  $\mathcal{A}_i = \{j \neq i : R_r \leq |\mathbf{r}_{ij}| < R_a\}$ , both sensed omnidirectionally. There is no attraction zone: spatial cohesion comes entirely from the density-dependent motility (the MIPS route), not from a Couzin cohesion term, and “Couzin” here labels only the zonal priority rule of repulsion over alignment over inertia:

$$\theta_i^\star(t+1) = \begin{cases} \arg\left(-\sum_{j \in \mathcal{R}_i} \mathbf{r}_{ij}/|\mathbf{r}_{ij}|\right), & \mathcal{R}_i \neq \emptyset, \\ \arg\left(\sum_{j \in \mathcal{A}_i} e^{i\theta_j(t)}\right), & \mathcal{R}_i = \emptyset \text{ and } \mathcal{A}_i \neq \emptyset, \\ \theta_i(t), & \mathcal{R}_i = \mathcal{A}_i = \emptyset. \end{cases} \quad (2)$$

Interactions run in  $O(N)$  on a cell-list partition.

The angular kick  $\xi_i$  is a symmetric  $\alpha$ -stable variate with a particle-dependent stability index,

$$\xi_i(t) \sim S_{\alpha_i(t)}(c = \eta, \beta_{\text{stab}} = 0, \mu = 0), \quad (3)$$

sampled with the Chambers–Mallows–Stuck representation [54] in the parametrisation of Samorodnitsky and Taqqu [55] and Nolan [56]. The Gaussian limit  $\alpha_i = 2$  is handled separately and recovers the original Vicsek noise. Since the kick is added to a heading and reduced modulo  $2\pi$ , the law that actually acts on the dynamics is the *wrapped* symmetric  $\alpha$ -stable distribution [48–50], whose mean resultant length  $\langle \cos \xi \rangle = \exp(-\eta^\alpha)$  we use throughout as the effective decorrelation scale; the consequences of that wrapping for the comparison between modes are developed in Sec. A.

## 2.2 The shared sigmoid

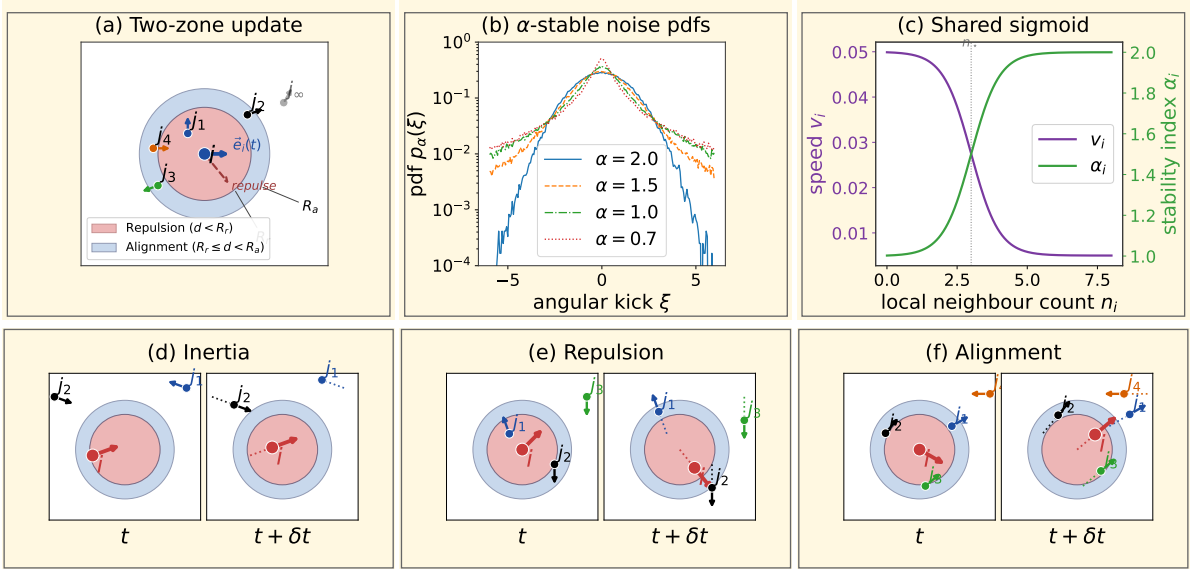
The single departure from canonical Vicsek dynamics is that both the speed and the noise stability index become local fields, controlled by the same neighbour count  $n_i(t) = |\mathcal{A}_i(t)|$  through one sigmoid,

$$\sigma_i(t) = \frac{1}{1 + \exp[-s(n_i(t) - n_\star)]}, \quad (4)$$

$$v_i(t) = v_{\max} - (v_{\max} - v_{\min}) \sigma_i(t), \quad (5)$$

$$\alpha_i(t) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \sigma_i(t). \quad (6)$$

A crowded particle ( $n_i \gg n_\star$ ) sits at  $\sigma_i \rightarrow 1$ ,  $v_i \rightarrow v_{\min}$ ,  $\alpha_i \rightarrow \alpha_{\max} = 2$ ; an isolated one ( $n_i \ll n_\star$ ) at  $\sigma_i \rightarrow 0$ ,  $v_i \rightarrow v_{\max}$ ,  $\alpha_i \rightarrow \alpha_{\min} = 1$ . The two channels are locked, slow meaning Gaussian and fast Cauchy (Fig. 1).



**Figure 1.** Vicsek–Couzin setup and update rules. (a) Two-zone geometry around a focal particle: repulsion disc ( $d < R_r$ ), alignment annulus ( $R_r \leq d < R_a$ ), full  $360^\circ$  vision, priority repulsion  $\rightarrow$  alignment  $\rightarrow$  inertia. (b) Symmetric  $\alpha$ -stable noise pdfs; smaller  $\alpha$  narrows the bulk and heavies the tails. (c) The shared sigmoid:  $n_i$  sets both  $v_i$  (purple) and  $\alpha_i$  (green), so a crowded particle is slow and Gaussian, an isolated one fast and Cauchy. (d)–(f) Worked inertia, repulsion and alignment steps of Eq. (2), each at  $t$  and  $t + \delta t$ . The shared sigmoid is the one departure from canonical Vicsek dynamics.

## 2.3 Reference and ablation modes

The two-feedback model (*double-adaptive*) runs against three controls on the same code path. Setting  $v_{\min} = v_{\max}$  and  $\alpha_{\min} = \alpha_{\max} = 2$  recovers the original Vicsek dynamics (*Vicsek reference*); setting instead  $\alpha_{\min} = \alpha_{\max} = 1$  gives the heavy-tailed *Cauchy reference*, the backdrop on top of which the feedbacks switch on. The *noise-shape only* mode activates the noise-shape channel with  $v_i$  frozen at  $v_{\max}$ ; the *motility only* mode activates the motility channel with  $\alpha_i$  frozen at the Cauchy value.

## 2.4 Order parameters and observables

The instantaneous polar order parameter is  $\varphi = \frac{1}{N} |\sum_i e^{i\theta_i}|$ , and the polar order parameter we report is its average,

$$\langle \varphi \rangle = \left\langle \frac{1}{N} \left| \sum_i e^{i\theta_i} \right| \right\rangle, \quad (7)$$

the susceptibility its scaled variance  $\chi = N \text{Var}(\varphi)$ , and the Binder cumulant  $U_4 = 1 - \langle \varphi^4 \rangle / (3 \langle \varphi^2 \rangle^2)$  [57], which we read with caution since every mode (the Cauchy reference included) drives it to the same 0.28–0.31 dip and on its own it does not diagnose the transition here. For spatial phase separation we use

$$s_{\text{sep}} \equiv \max_{\text{bins}} \frac{n_{\text{bin}}}{\text{med}(n_{\text{bin}})}, \quad (8)$$

the ratio of the densest grid cell to the median cell on a  $10 \times 10$  partition: a homogeneous fluid sits at  $s_{\text{sep}} \simeq 1$ , a phase-separated configuration well above, and a grid-free pair-correlation analogue (the 95th-percentile / median ratio of the per-particle local density within radius  $R = 1$ ) ranks the modes the same way. To separate a traveling band from an isotropic patch we use the band-soliton index,

$$b \equiv \frac{\max_{x_{\parallel}} \rho(x_{\parallel})}{\langle \rho \rangle}, \quad (9)$$

the maximum of the time-averaged density profile along the instantaneous polar-flow axis  $x_{\parallel}$  divided by the spatial mean. A metric-Vicsek traveling band gives  $b \gtrsim 1.5$ ; an isotropic patch gives  $b \simeq 1$ .

## 2.5 Numerical protocol

The finite-size scan spans nine sizes  $L \in \{15, 22, 30, 45, 64, 90, 128, 180, 256\}$  at fixed density  $\rho_0 = N/L^2 = 2.22$  (up to  $N \simeq 1.5 \times 10^5$  particles), with 1500 warm-up and 1000 measurement steps from  $\theta_i \equiv 0$ . The load-bearing slopes (Table 2 and the  $\chi_{\text{max}}(L)$  panel of Fig. 3) come from a homogeneous protocol with ten seeds at *every* size, so the bootstrap intervals rest on a uniform replication budget, while the  $\eta$ -resolved diagnostic curves (the remaining panels of Fig. 3) use three seeds per (mode,  $L, \eta$ ) cell. To separate a robust phase-separation split from a parameter-window effect we repeat the scan at  $\rho_0 \in \{1.0, 3.0\}$  (sizes 22 to 128). The noise grid is  $\eta \in \{0.005, 0.010, 0.020, 0.035, 0.050, 0.075, 0.100, 0.150, 0.200, 0.300\}$ , and the fixed parameters are  $R_r = 0.5$ ,  $R_a = 0.7$ ,  $n_{\star} = 3$ ,  $s = 2$ ,  $v_{\text{min}} = 0.005$ ,  $v_{\text{max}} = 0.05$ ,  $\alpha_{\text{min}} = 1$ ,  $\alpha_{\text{max}} = 2$ . The interaction loop is numba-JIT compiled. A small- $\eta$  refinement  $\{0.001, 0.002, 0.003, 0.0075\}$  at the first five sizes (two seeds) confirms that  $\chi_{\text{max}}$  does not lurk outside the original window, bar the noise-shape-only ablation at  $L = 45$ , which we report with its updated value.

## 3 Results

A visual reference fixes intuition before the diagnostics. We compare motility-only and double-adaptive configurations at  $\rho_0 = 2.22$  across three sizes  $L \in \{30, 90, 128\}$  and three noise levels spanning the ordered ( $\eta = 0.020$ ), near-critical ( $\eta = 0.10$ ) and disordered ( $\eta = 0.30$ ) regimes (Fig. 2). Deep in the ordered phase the flock is almost fully aligned in both modes; at near-critical  $\eta$  it breaks into locally ordered slow patches in a faster disordered gas; at  $\eta = 0.30$  the particles fill the box with no coherent heading. The near-critical column is the operating point at which the rectification is active, and there the double-adaptive panels hold a tighter dense core than their motility-only neighbours at every size.

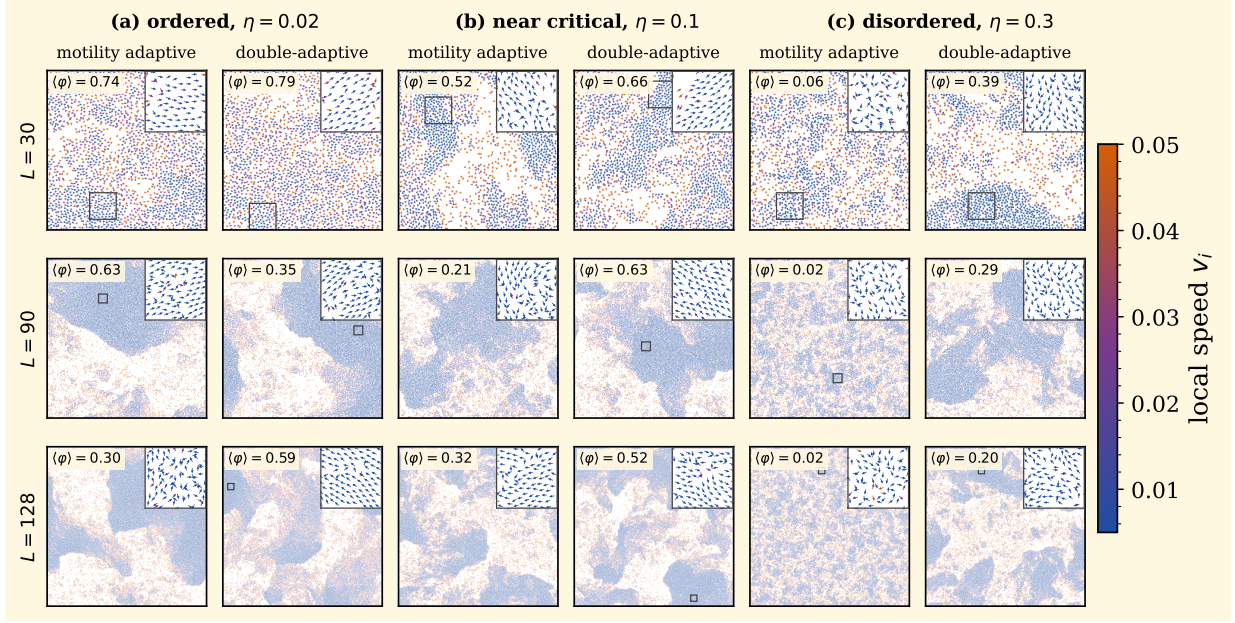
### 3.1 Global diagnostics

#### 3.1.1 Finite-size scaling of the susceptibility

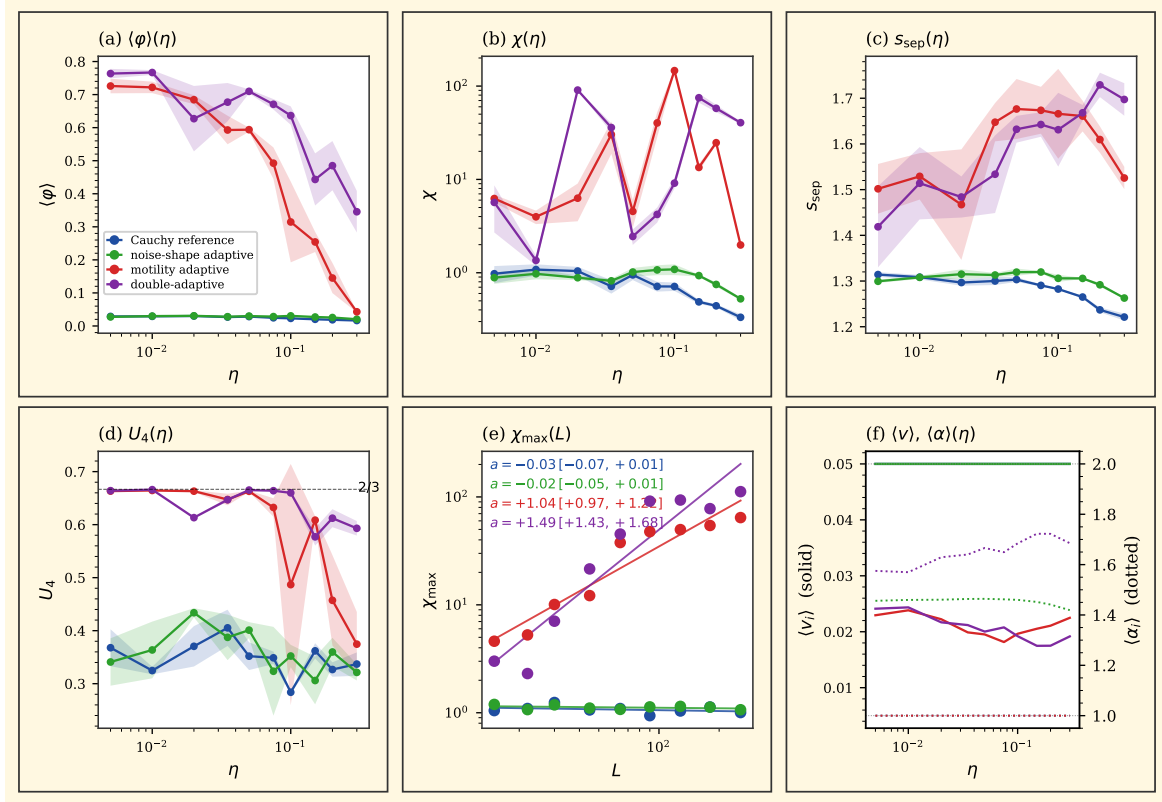
To see whether the two adaptations leave a thermodynamic trace we run all four heavy-tailed modes on the nine-size grid of Sec. II E and read the susceptibility peak  $\chi_{\text{max}}(L)$ , its slope  $a$  in  $\chi_{\text{max}} \sim L^a$  separating them (Fig. 3, Table 2).

The slopes split cleanly into two regimes. The three fixed-noise-shape modes (Vicsek–Gaussian, Cauchy, noise-shape only) carry slopes indistinguishable from zero ( $-0.03$ ,  $-0.03$ ,  $-0.02$ ), every





**Figure 2.** Real-space configurations at  $\rho_0 = 2.22$ , single seed, after 3000 warm-up steps, three sizes  $L \in \{30, 90, 128\}$  (one row each). Per row the noise runs left to right: (a)  $\eta = 0.020$  (ordered), (b)  $\eta = 0.10$  (near-critical), (c)  $\eta = 0.30$  (disordered); each placing motility-only left of its double-adaptive counterpart. Arrows are coloured by local speed  $v_i$  (shared colorbar), a deliberate departure from a geometry-only snapshot because the speed field is the control variable, with a  $5 \times 5$  zoom inset and the polar order  $\langle \varphi \rangle$  per panel.



**Figure 3.** FSS diagnostics, four modes: Cauchy reference (blue), noise-shape only (green), motility only (red), double-adaptive (purple). The  $\eta$ -resolved panels (three seeds) show, versus  $\eta$  at the largest pilot size  $L = 128$ : (a) polar order  $\langle \varphi \rangle$ , (b) susceptibility  $\chi$ , (c)  $s_{\text{sep}}$ , (d) Binder cumulant  $U_4$ ; (e) log-log  $\chi_{\text{max}}$  versus  $L$  with fitted slopes, over the nine homogeneous sizes ( $L = 15$  to  $256$ , ten seeds) of Table 2; (f) mean speed  $\langle v_i \rangle$  (solid) and mean stability  $\langle \alpha_i \rangle$  (dotted).

95% interval straddling the origin ( $[-0.05, +0.01]$ ,  $[-0.07, +0.01]$ ,  $[-0.05, +0.01]$ ): none grows a size-dependent susceptibility on the nine-point lever, the Cauchy-reference  $\chi_{\max}$  staying flat near unity. The two motility-active modes do grow one:  $a_{\text{motility}} = +1.04$  ( $[+0.97, +1.22]$ ) and  $a_{\text{full}} = +1.49$  ( $[+1.43, +1.68]$ ), the full mode steeper and the intervals disjoint. These are growth rates of a phase-separation susceptibility, not critical exponents: extending the lever from seven to nine sizes lowers both slopes ( $a_{\text{motility}} : +1.29 \rightarrow +1.04$ ,  $a_{\text{full}} : +1.95 \rightarrow +1.49$ ), so the apparent exponent is not stable, and a single phase-separated domain would saturate at the  $d = 2$  ceiling  $\chi_{\max} \sim N$ ; we find neither a data collapse nor the  $U_4$  crossing a genuine critical point would require. The synergy diagnostic

$$\Delta \equiv (a_{\text{full}} - a_{\text{Cauchy}}) - (a_{\text{noise}} - a_{\text{Cauchy}}) - (a_{\text{motility}} - a_{\text{Cauchy}}) \quad (10)$$

vanishes for two linearly-combining channels above the Cauchy backdrop. With the Cauchy and noise-shape slopes both essentially zero,  $\Delta_n$  (on the first  $n$  sizes) reduces to  $a_{\text{full}} - a_{\text{motility}} = +1.49 - (+1.04)$ , and the full nine-size fit returns  $\Delta_9 = +0.45$ , 95% seed-bootstrap interval  $[+0.31, +0.60]$ , each mode taken at its own near-critical noise (and robust to a size-jackknife, below).

This apparent excess is a noise-level effect, not a cooperation between the channels. The full mode's dense kick ( $\alpha \rightarrow 2$ ) decorrelates more slowly than the motility-only mode's heavy-tailed kick ( $\alpha = 1$ ) at the same  $\eta$ , with  $\Gamma(\eta, \alpha) = 1 - e^{-\eta^\alpha}$  (Sec. A). Retuning motility-only over a matched grid  $\eta \rightarrow \eta^2$ , so its  $\alpha = 1$  kick decorrelates as fast as the full mode's  $\alpha = 2$  kick, collapses the excess on the same nine sizes:  $a_{\text{motility}}$  rises from  $+1.04$  to  $+1.47$  ( $[+1.41, +1.74]$ ), level with  $a_{\text{full}} = +1.49$ , so  $\Delta_9 \rightarrow +0.02$ , additive within error. The same control runs through the microscopic diagnostics below (Table 1), where the dense  $g(r)$  and  $C(\tau)$  gaps do not merely vanish but reverse sign, because the global  $\eta \rightarrow \eta^2$  retune also quiets the dilute phase. The faster  $\chi_{\max}$  growth, the sharper dense  $g(r)$  and the slower dense  $C(\tau)$  are thus one effect, the lower dense effective noise of the rectified kick made visible; because the retune lowers noise everywhere, this bounds it as a noise effect rather than isolating the dense phase on its own.

**Table 1.** Full-mode advantage at each mode's own near-critical noise (left) and under the matched- $\Gamma$  retune of motility-only (right), for the three load-bearing observables. Equalising the dense decorrelation removes the susceptibility excess (additive within error) and reverses the static and dynamic gaps, the global retune over-quieting the dilute phase, so all three are one dense-noise effect rather than a cooperation.

| Full – motility                | each-own- $\eta$ | matched $\Gamma$ |
|--------------------------------|------------------|------------------|
| $\chi_{\max}$ slope $\Delta_9$ | +0.45            | +0.02            |
| dense $g(r \simeq 0.6)$ gap    | +0.153           | −0.07            |
| dense $C(\tau)$ gap            | +0.13 to +0.24   | −0.08 to −0.42   |

Figure S3 makes the point directly: the local slope  $a_{\text{eff}} = d \ln \chi_{\max} / d \ln L$  drifts rather than locking to a constant, and the susceptibility peak  $\eta^*(L)$  migrates toward the small- $\eta$  boundary rather than to a fixed  $\eta_c$ , so a  $(\eta - \eta_c)L^{1/\nu}$  collapse cannot be defined. The load-bearing evidence lies in the microscopic signature below.

The cumulative budget is positive at every truncation and excludes zero from  $n = 5$  onward. The motility-only  $\chi_{\max}(L)$  rises monotonically across all nine sizes; the full-mode series is monotone through  $L = 128$  but scatters at the two largest sizes ( $93.6 \rightarrow 77.9 \rightarrow 112.0$  at  $L = 128, 180, 256$ ), the pooled estimator staying seed-sensitive at large  $L$  under heavy tails (ten seeds being the budget we could afford at  $N \simeq 1.5 \times 10^5$ ). The slope CIs in Table 2 bootstrap over seeds at fixed  $L$  and so report seed scatter, not this across-size instability: a size-jackknife moves  $a_{\text{full}}$  to  $+1.67$  (dropping  $L = 256$ ) or  $+1.57$  (dropping  $L = 180$ ) and  $a_{\text{motility}}$  to  $+1.15$  or  $+1.07$ , so  $\Delta = a_{\text{full}} - a_{\text{motility}}$  stays in  $[+0.45, +0.52]$  under either deletion. The each-own- $\eta$  excess therefore does not hinge on the non-monotone tail, and its noise-level origin is established above; the microscopic diagnostics below carry the load-bearing evidence.



**Table 2.** Susceptibility peaks  $\chi_{\max}(L)$  on the nine-size grid ( $L = 15$  to 256) and log-log FSS slopes  $a$  for the Vicsek–Gaussian reference and the four heavy-tailed modes, ten seeds per size (homogeneous protocol). The 95% slope CI is bootstrapped over the ten seeds ( $B = 10\,000$  resamples) [58]. The three fixed-noise-shape modes stay flat within  $\pm 0.2$  across the full size range; only the two motility-active modes grow a size-dependent  $\chi_{\max}$ , the double-adaptive slope the steeper, and adding  $L = 180, 256$  lowers both motility-active slopes (the apparent exponent is not stable).

| Mode                 | $L = 15$ | $L = 22$ | $L = 30$ | $L = 45$ | $L = 64$ | $L = 90$ | $L = 128$ | $L = 180$ | $L = 256$ | slope $a$ | 95% CI         |
|----------------------|----------|----------|----------|----------|----------|----------|-----------|-----------|-----------|-----------|----------------|
| Vicsek (Gaussian)    | 1.13     | 1.18     | 1.07     | 1.06     | 1.10     | 1.08     | 1.02      | 1.06      | 1.06      | −0.03     | [−0.05, +0.01] |
| Cauchy reference     | 1.05     | 1.09     | 1.24     | 1.06     | 1.09     | 0.94     | 1.04      | 1.13      | 1.00      | −0.03     | [−0.07, +0.01] |
| noise-shape adaptive | 1.19     | 1.07     | 1.18     | 1.11     | 1.07     | 1.13     | 1.15      | 1.13      | 1.07      | −0.02     | [−0.05, +0.01] |
| motility adaptive    | 4.60     | 5.26     | 10.07    | 12.18    | 37.77    | 47.66    | 49.81     | 54.36     | 64.35     | +1.04     | [+0.97, +1.22] |
| double-adaptive      | 3.01     | 2.31     | 7.05     | 21.52    | 45.27    | 91.40    | 93.58     | 77.91     | 111.96    | +1.49     | [+1.43, +1.68] |

A single working density could be doing the work here, so we repeated the scan at  $\rho_0 = 1.0$  and 3.0 (Table S1, Supplementary Material). The three fixed-noise modes stay flat there as well ( $|a| \leq 0.09$ ) while both motility-active modes grow at every density, so the split is set by the motility-driven phase separation and not by a tuned parameter window. The density trend inside that scan is itself informative. At  $\rho_0 = 1.0$  the full and motility-only slopes coincide (+0.63 against +0.64) and the excess vanishes, whereas it reaches  $\simeq +0.8$  at  $\rho_0 = 2.22$  and +1.0 at  $\rho_0 = 3.0$ , which is what one expects if the rectification acts inside a dense phase that the motility channel has to build first and that grows with density.

The reading is mechanistic. Inside a dense patch the alignment sum over many annulus neighbours is dominated by typical kicks, so the Cauchy bursts of the heavy-tail-only modes wash out before they can drive any  $\chi_{\max}$  growth, while the motility channel sidesteps that averaging and the noise-shape rectification adds on top because  $\alpha_i \rightarrow 2$  damps the residual heading fluctuations in the same patch, giving the  $\Delta a \simeq +0.4$  excess. The  $U_4$  panel of Fig. 3d adds nothing here: all four modes, the Cauchy reference included, reach the same  $U_4 \rightarrow 0.28\text{--}0.31$  dip at  $L = 64$ , Cauchy fluctuations of the global order in the dilute phase pushing  $U_4$  down with no two-phase coexistence, so the dip is not a reliable first-order signature and we read the spatial structure from  $s_{\text{sep}}$  instead.

### 3.1.2 Density separation and real-space contrast

The next question is whether the dense regions are spatially separated. On the same protocol, at the  $\chi$ -peak  $\eta$  and the three largest sizes (Fig. 3c), the two Cauchy-only modes are essentially homogeneous ( $s_{\text{sep}} \simeq 1.2$ , the binning floor),  $s_{\text{sep}}$  in fact falling slowly with  $L$ . Only the two motility-active modes phase separate, both plateauing near  $s_{\text{sep}} \simeq 1.7$ , the double-adaptive model above the motility-only ablation by +0.24 at  $L = 64$  on these three-seed pilot curves (the dedicated ten-seed paired gap of Table S2 reads +0.165 there) and closing to +0.01 by  $L = 128$ , so  $s_{\text{sep}}$  reads the same split as the susceptibility slope, the noise-shape surplus largest at intermediate sizes before the plateau closes it.

The snapshots (Fig. 2, top row,  $L = 30$ ) make the contrast concrete. The two modes are nearly indistinguishable at  $\eta = 0.020$  ( $\langle\varphi\rangle = 0.74$  motility-only, 0.79 full); at near-critical  $\eta = 0.10$  the double-adaptive model already holds more order (0.52 against 0.66); and at  $\eta = 0.30$  the motility ablation collapses to  $\langle\varphi\rangle = 0.06$  while the double-adaptive model still holds 0.39, its dense slow patches staying coherently aligned where the motility-only patches scramble. Inside the dense phase  $\langle\alpha_i\rangle$  reaches 1.5–1.6 for the full mode and damps the Lévy bursts that would otherwise reshuffle the slow particles, whereas motility-only takes the full Cauchy kick there at every step.

## 3.2 Parametric analyses and distributions

### 3.2.1 Robustness across the feedback parameters

Everything above rests on one working point,  $(n_*, s) = (3, 2)$ , and on the nine sizes of the FSS grid. Two parametric checks, reported in full in the Supplementary Material, ask whether the result hangs on either. Sweeping the two sigmoid parameters on a  $13 \times 13$  grid at  $L \in \{30, 90, 128\}$  spreads the cohesion gain across the whole MIPS-active corner  $n_* \lesssim 4$  (Fig. S4) instead of confining it to a knife-edge cell, and kills it for  $n_* \geq 5$ , where the threshold sits past the typical occupancy  $\bar{n}_i \simeq 1.7$  and neither feedback engages. The two observables do not optimise on the same cell, however:  $\chi_{\max}$  peaks where the rectification fires almost everywhere and washes out the dense–dilute asymmetry that  $s_{\text{sep}}$  needs. That mismatch is the first sign that they are not reading one cooperative effect. Filling in the  $L$  axis between the FSS nodes (Fig. S5) then leaves the  $s_{\text{sep}}$  gap positive but modest, and a ten-seed paired test resolves it only at  $L = 64$  ( $+0.165$ , 95% seed-bootstrap CI  $[+0.109, +0.223]$ , Wilcoxon  $p = 0.002$ ), returning nothing distinguishable from zero at  $L = 50$  and  $80$  (Table S2). Like the susceptibility excess, it is the noise-shape channel reading out at each mode’s own noise.

### 3.2.2 Transition order, dense-phase geometry, and the Vicsek baseline

Two further checks, detailed in the Supplementary Material, fix the order of the transition and the geometry of the dense phase. The order-parameter distribution  $P(\langle \varphi \rangle)$  stays unimodal at every size and the quasi-static  $\eta$  ramp opens no hysteresis loop (loop area  $\leq 0.006$ ), so the transition is continuous, or at most very weakly first-order (Fig. S1). The time-averaged density profile along the polar-flow axis is flat for all modes (band-soliton index  $b \simeq 1$ ), making the motility-built dense phase an isotropic comoving patch, not a travelling band (Fig. S2).

The four heavy-tailed modes share a Cauchy backdrop and compare directly; the canonical Vicsek–Gaussian baseline sits outside. We recover it from Eqs. (1)–(6) with  $v_{\min} = v_{\max} = 0.05$  and  $\alpha_{\min} = \alpha_{\max} = 2$  on the same nine-size grid (its row in Table 2). It develops no critical  $\chi_{\max}(L)$  scaling at our density: the peak stays in 1.0–1.2, slope  $a_{\text{Vicsek}} = -0.03$  ( $[-0.05, +0.01]$ ), the metric-Vicsek susceptibility damped below the resolution of our lever. The spatial diagnostic separates more sharply:  $s_{\text{sep}} = 1.24$  for Vicsek at  $L = 64$ , nearly identical to Cauchy and noise-shape (1.25) and within the homogeneous-fluid range Cates–Tailleur theory anticipates for a fixed-speed model, while the full mode lands at 1.72. The Binder cumulants stay similar across all five modes ( $U_4 \in [0.3, 0.5]$ ).

## 3.3 Local diagnostics

### 3.3.1 Dense-phase structure

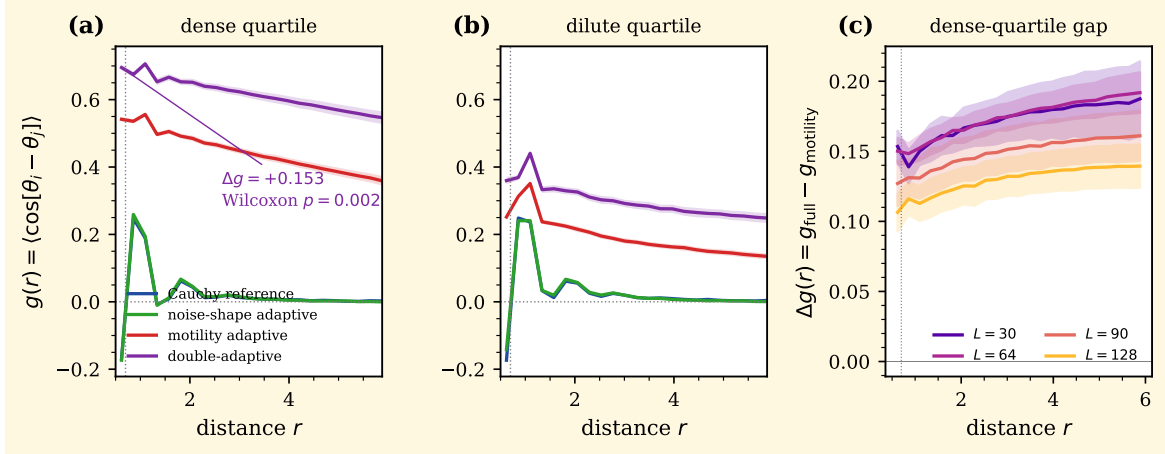
The susceptibility slope and  $s_{\text{sep}}$  both say the dense phase is more pronounced under the full mode; neither says how it is organised. We therefore label the connected dense components of the coarse-grained density field at  $L = 30$  over ten seeds and compare the two motility-active modes patch by patch, with the clustering protocol and the number-fluctuation fit given in full in the Supplementary Material (Figs. S6, S7). Both modes fragment the box into  $\sim 160$ – $190$  dense clusters, but the full mode pushes the largest one from  $55.0 \pm 7.5$  to  $176.6 \pm 42.3$  particles, a paired gap of  $+121.6$  ( $[+40.5, +206.6]$ , Wilcoxon  $p = 0.02$ ), while the cluster count carries no significant signal ( $p = 0.19$ ). The rectification fuses dense patches into a few large droplets at conserved dense area; it does not nucleate new ones. Number fluctuations take the argument only half way, confirming the phase separation ( $\zeta = 0.79$  and  $0.82$  against  $\simeq 0.57$  for the fixed-speed references) without separating the two motility-active modes from each other, which is why the mechanism has to be read off the dense-quartile observables below.

### 3.3.2 Heading correlation $g(r)$ in dense and dilute regions

If the rectification acts in the dense phase, its fingerprint should show up independently of cluster geometry. We measure the density-stratified heading correlation

$$g(r) = \langle \cos[\theta_i(t) - \theta_j(t)] \rangle_{|\mathbf{x}_i - \mathbf{x}_j| \in [r, r + \Delta r]} \quad (11)$$

for the four heavy-tailed modes at near-critical  $\eta$ ,  $L = 30$ , ten seeds, six snapshots per seed, the reference particle  $i$  restricted to a density quartile of the local-density field  $\rho_i = \#\{j : |\mathbf{x}_j - \mathbf{x}_i| < 1\}$  and partner  $j$  ranging over the whole box (Fig. 4).



**Figure 4.** Density-stratified heading correlation  $g(r)$  at near-critical  $\eta$ ,  $L = 30$ , ten seeds. (a) dense and (b) dilute quartile for the four modes, with seed-level error bands; (c) dense-quartile gap  $\Delta g(r)$  between double-adaptive and motility-only over  $r \in [0.5, 6]$  at  $L \in \{30, 64, 90, 128\}$ .

The dense-phase correlation splits the modes into two groups. The Cauchy and noise-shape modes reach  $g(r \simeq 0.6) = -0.169 \pm 0.003$  and  $-0.173 \pm 0.004$  in the dense quartile, both negative: heavy-tailed bursts actively anti-align particles inside the dense patches. Motility-only reaches  $0.542 \pm 0.007$ , the full mode  $0.695 \pm 0.008$ , a paired  $\Delta g = +0.153$  (95% seed-bootstrap CI  $[+0.133, +0.175]$ , Wilcoxon  $p = 0.002$ ). The dilute-phase correlations barely move with the noise-shape adaptation, so the asymmetry is the direct microscopic expression of the rectification.

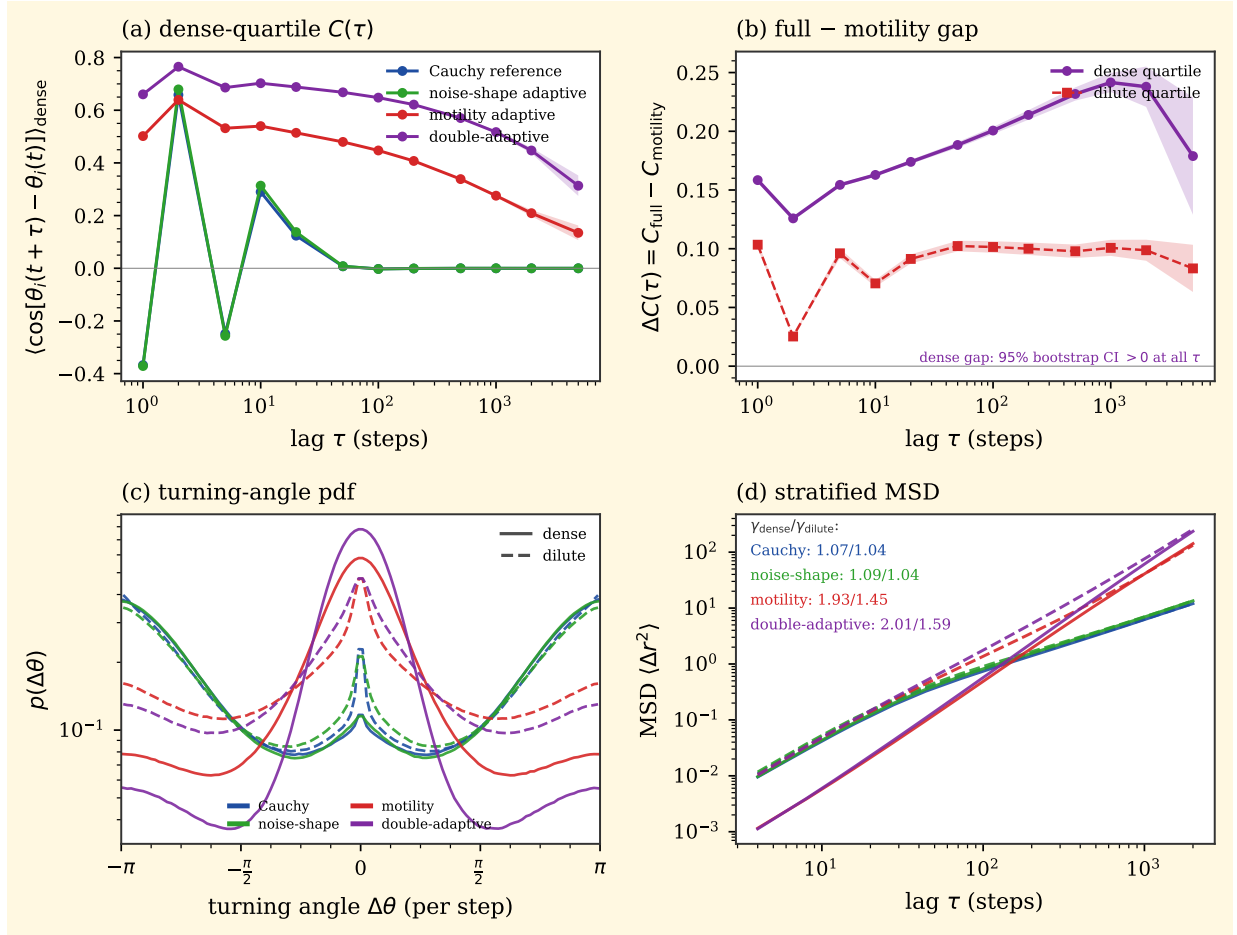
The signature survives the FSS range. A ten-seed test holds the dense-quartile gap in  $\Delta g(r \simeq 0.6) \in [+0.11, +0.15]$  across  $L = 64$  to  $128$  (each 95% seed-bootstrap CI excluding zero, Wilcoxon  $p = 0.002$ ), narrowing mildly from  $+0.15$  at  $L \leq 64$  to  $+0.11$  at  $L = 128$ . Over the full  $r \in [0.5, 6]$  profile (Fig. 4c) it does not decay with  $r$ : the outer-range mean ( $r > 3$ ,  $+0.14$  to  $+0.18$ ) sits if anything above the inner-range mean ( $r \in (0.5, 1.5)$ ,  $+0.11$  to  $+0.15$ ), larger beyond the alignment cutoff  $R_a = 0.7$  than inside it, so the rectification imposes a coherence that outlasts the metric interaction range.

This gap is the rectification of the dense noise made visible, not a coherence the full mode would hold beyond it. With the effective decorrelation  $\Gamma(\eta, \alpha) = 1 - e^{-\eta^\alpha}$  of the appendix, the full mode's dense kick ( $\alpha \rightarrow 2$ ,  $\eta = 0.15$ ) decorrelates at  $\Gamma = 0.022$ , four times slower than the motility-only dense kick ( $\alpha = 1$ ,  $\eta = 0.10$ ,  $\Gamma = 0.095$ ). Retuning motility-only to  $\eta = 0.0225$ , where its  $\alpha = 1$  kick matches the full mode's dense  $\Gamma$ , reverses the gap to  $\Delta g(r \simeq 0.6) = -0.07$  (95% seed-bootstrap CI  $[-0.07, -0.06]$ , Wilcoxon  $p = 0.002$ ), the retuned motility phase now marginally the sharper. The dense correlation gap therefore measures the lower effective noise the noise-shape feedback installs in the dense phase, not an alignment held at equal noise.

### 3.3.3 Single-particle dynamics: $C(\tau)$ , turning angles and diffusion

The dense  $g(r)$  measures the static sharpness of local alignment but says nothing about how a heading evolves. Three single-particle diagnostics, all on the same near-critical runs at  $L = 30$ ,  $\rho_0 =$

2.22, close that gap (Fig. 5). Persistence comes first. The appendix (Sec. A) predicts a rotational decorrelation rate  $\Gamma(\rho, \eta) \simeq \eta^{\alpha(\rho)}$ , a factor  $\eta^{-1} \simeq 10$  faster in the dilute ( $\alpha \rightarrow 1$ ) than the dense ( $\alpha \rightarrow 2$ ) phase at our working point. For each mode at its near-critical  $\eta$ , ten seeds,  $n_{\text{meas}} = 20\,000$  steps, we tag every particle every 50 steps by quartile of the local count and compute  $C_{\text{dense}}(\tau) = \langle \cos[\theta_i(t + \tau) - \theta_i(t)] \rangle_{i \in \text{dense}}$  on lags  $\tau \in \{1, 2, 5, 10, 20, 50, 100, 200, 500, 1\,000, 2\,000, 5\,000\}$  (and  $C_{\text{dilute}}$  likewise). The four modes order the same way at every lag (Fig. 5a): Cauchy and noise-shape drop fastest, motility-only sits above, and the full model above it by a nearly constant offset. The gap  $\Delta C(\tau) = C_{\text{full}}(\tau) - C_{\text{motility}}(\tau)$  stays in the band  $+0.13$  to  $+0.24$  over  $\tau \in [1, 5\,000]$  in the dense quartile, its bootstrap 95% CIs (ten seeds,  $B = 10\,000$  resamples) excluding zero throughout; with  $\sim 4 \times 10^7$  sample pairs per mode we quote bootstrap CIs rather than a between-seed  $z$ , which saturates beyond a few sigmas. The dilute gap is a few percent and its CI brackets zero past  $\tau \simeq 100$ . The matched- $\Gamma$  control carries over to the dynamics: retuning motility-only to  $\eta = 0.0225$  reverses the dense gap at every lag, from  $\Delta C = -0.08$  at  $\tau = 1$  to  $-0.42$  at  $\tau = 5\,000$ , so the dynamic dense-coherence advantage, like the static one, is the visible signature of the rectified dense noise, not coherence beyond it.



**Figure 5.** Single-particle dynamics at  $L = 30$ ,  $\rho_0 = 2.22$ , near-critical  $\eta$ . Top row (ten seeds, 20 000 steps): (a)  $C_{\text{dense}}(\tau)$  per mode ( $\pm 1$  SE bands); (b) the gap  $\Delta C(\tau) = C_{\text{full}} - C_{\text{motility}}$  in the dense and dilute quartiles, the dense gap carrying a 95% seed-bootstrap CI above zero at every lag. Bottom row, per-particle increments (six seeds; hue keys the mode, line style the density class, solid dense, dashed dilute): (c) turning-angle pdf  $p(\Delta\theta)$ ; (d) density-conditioned MSD  $\langle \Delta r^2 \rangle(\tau)$  with fitted  $\gamma_{\text{dense}}/\gamma_{\text{dilute}}$ . The dense  $C(\tau)$  gap persists far beyond the polar-order correlation time; only the double-adaptive increment turns Gaussian-narrow in the dense phase ( $\alpha \rightarrow 2$ ), and that dense phase is ballistic ( $\gamma \simeq 2$ ) while the dilute gas is super-diffusive but sub-ballistic.

The dense-phase plateau matches the  $\Gamma \simeq \eta^{\alpha(\rho)}$  form of Sec. A.1: a more persistent heading once  $\alpha$  switches from 1 to 2 inside the motility-built dense phase. Measured at a single  $\eta$  per

mode and flat across lags, the gap probes the dense-phase  $\alpha \rightarrow 2$  conversion, not the  $\eta$ -exponent of  $\Gamma$  itself, which an  $\eta$ -resolved run would test, extending the static dense  $g(r)$  of Fig. 4c to the single-particle dynamic well beyond the polar-order correlation time.

The per-step turning-angle distribution exposes what the persistence plateau rests on: the noise gate is a switch in the reorientation statistics. Stratifying the same runs by density quartile (six seeds, 8000 steps; Fig. 5c), in the dilute gas both motility-active models turn with a heavy, Cauchy-like tail (weight beyond  $|\Delta\theta| > \pi/2$  of 0.34 double-adaptive, 0.41 motility-only). In the dense phase they part company: the double-adaptive increment collapses onto a Gaussian-like core (tail weight 0.14, circular variance 0.34), whereas motility-only stays heavy (tail weight 0.22, circular variance 0.50) with  $\alpha$  pinned at 1. The gate cuts the dense tail weight by 2.4 for the double-adaptive model against 1.9 for motility-only, so the  $C(\tau)$  plateau, read at the level of increments, is a switch from heavy-tailed to Gaussian turning inside the dense phase. The two fixed-speed references keep a heavy dense-quartile tail, so the Gaussian conversion needs the motility channel to build the dense phase first.

More persistent headings ought to show up as straighter trajectories, and the density-conditioned mean squared displacement, reported in the Supplementary Material, confirms that they do. The dense phase is ballistic ( $\gamma_{\text{dense}} = 2.01 \pm 0.01$  double-adaptive, 1.93 motility-only) while the dilute gas is super-diffusive but sub-ballistic. The Gaussian gate makes the dense droplets advance as cleaner ballistic units; it does not change the mechanism read above.

### 3.3.4 Decoupled sigmoids: a four-way anatomy of the rectification

The shared sigmoid cannot say which ingredient does the work. We split it into two independent profiles and run four tests: the threshold plane, the density dependence, the alignment kernel and the patch persistence.

We replace the shared sigmoid by two independent ones with parameters  $(n_\star^v, s^v)$  and  $(n_\star^\alpha, s^\alpha)$ , setting both thresholds to 3 and both slopes to 2 to recover the shared model. Sweeping the quadrant  $(n_\star^v, n_\star^\alpha) \in \{1, 2, 3, 5, 8\}^2$  at fixed  $s^v = s^\alpha = 2$ , ten seeds per cell at  $L = 30$ ,  $\eta = 0.15$ , we record the seed-paired gap  $\Delta g(r \simeq 0.6) = g_{\text{full}}^{\text{decoupled}} - g_{\text{motility}}$  (Fig. 6).

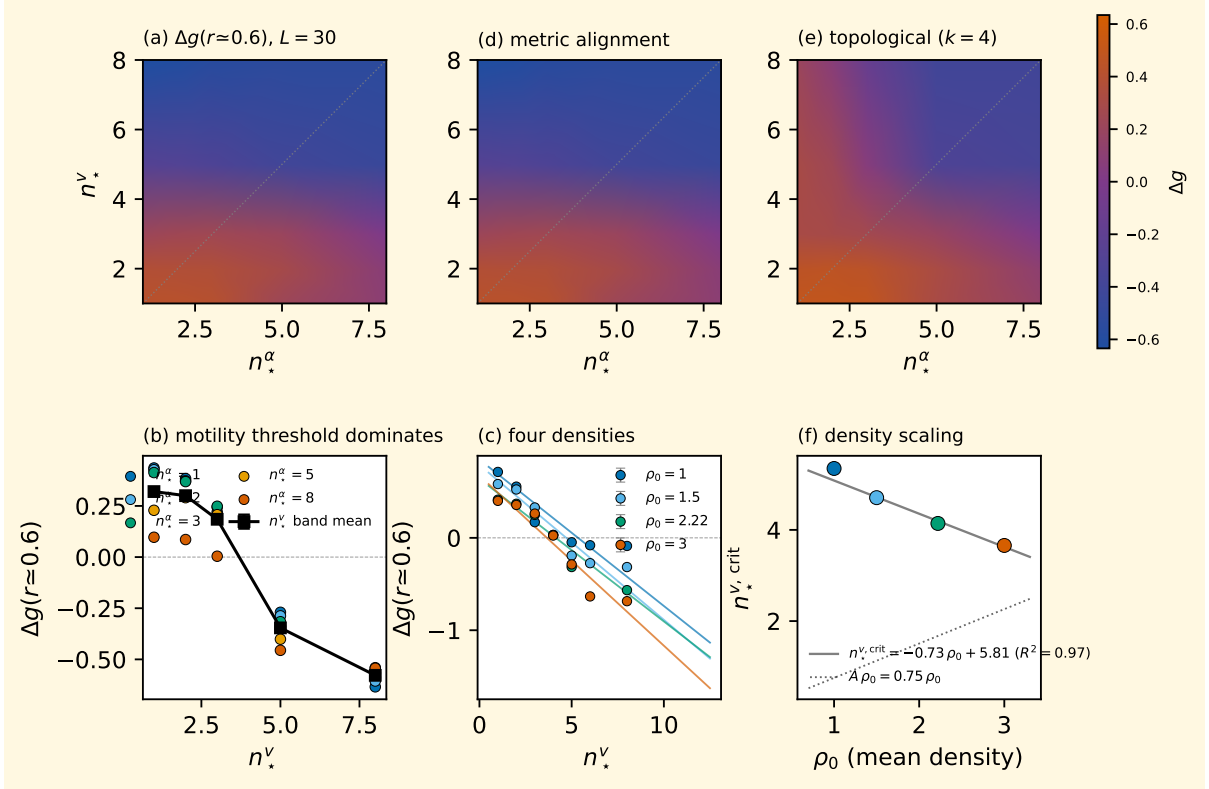
The map splits the quadrant into two horizontal bands, separated not by the diagonal but by a critical motility threshold  $n_\star^v \simeq 4$ , just above the typical alignment-zone occupancy  $\rho_0 \pi (R_a^2 - R_r^2) \simeq 1.7$ . Averaging each  $n_\star^v$  band over  $n_\star^\alpha$ , the mean gap steps monotonically from  $+0.32$  (95% seed-bootstrap CI  $[+0.31, +0.33]$ ) at  $n_\star^v = 1$  down to  $-0.58$  ( $[-0.59, -0.57]$ ) at 8, the adjacent bands non-overlapping and the sign flipping at  $n_\star^v \simeq 4$ , while the matching  $n_\star^\alpha$  bands stay near zero ( $+0.03$  for  $n_\star^\alpha \leq 3$ , down to  $-0.16$  at  $n_\star^\alpha = 8$ ): the noise-shape threshold is passive. The shared-sigmoid diagonal ( $n_\star^v = n_\star^\alpha$ ) crosses straight from the positive bands at  $n_\star^v \leq 3$  to the negative bands at  $n_\star^v \geq 5$ , so the sign is set by  $n_\star^v$  alone, not by the timing offset, as a linear fit confirms ( $\Delta g \simeq -0.144 n_\star^v + 0.52$ ,  $R^2 = 0.88$ , against  $R^2 = 0.04$  on  $n_\star^\alpha$  and 0.28 on the timing offset).

The mechanism is simple. At  $n_\star^v = 3$ , just above typical occupancy, dense-patch particles fire the motility channel while the rest saturate at  $v_{\text{max}}$ . Push  $n_\star^v$  to 5 or 8 and the motility channel almost never fires: the decoupled model collapses onto the noise-shape-only ablation, which has  $g(r \simeq 0.6) = -0.17$  at  $L = 30$  (Sec. 3.3.2), so its gap with motility-only is large and negative, the lower band. Drop  $n_\star^v$  below 3 and the motility channel engages aggressively, dense patches grow, and the noise-shape rectification adds coherence on top whether it triggers early or late, the upper band. The slowdown is again the precondition.

How does the critical motility threshold  $n_\star^{v,\text{crit}}$  move with the global density  $\rho_0$ ? The mean-density reading takes the relevant count to be  $A\rho_0 = \pi(R_a^2 - R_r^2)\rho_0$  and predicts  $n_\star^{v,\text{crit}} \propto \rho_0$  with slope  $A \simeq +0.75$ . Sweeping  $n_\star^v \in \{1, 2, 3, 4, 5, 6, 8\}$  at  $\rho_0 \in \{1.0, 1.5, 2.22, 3.0\}$ ,  $n_\star^\alpha = 3$  fixed, ten seeds, we read the zero crossing of  $\Delta g(n_\star^v) \simeq a(\rho_0) n_\star^v + b(\rho_0)$ . Per-density fits all reach  $R^2 \geq 0.86$  (Fig. 6c) and

$$n_\star^{v,\text{crit}}(\rho_0) = -0.73 \rho_0 + 5.81, \quad R^2 = 0.97. \quad (12)$$





**Figure 6.** Decoupled-sigmoid dissection of the gap  $\Delta g(r \approx 0.6) = g_{\text{full}}^{\text{decoupled}} - g_{\text{motility}}$  at  $L = 30$ , ten seeds per cell. (a) Heatmap on the  $(n_*^\alpha, n_*^v)$  plane (dashed  $n_*^v = n_*^\alpha$ ). (b)  $\Delta g$  vs  $n_*^v$  keyed by  $n_*^\alpha$ , with band means ( $\pm 95\%$  CI) crossing zero at  $n_*^v \approx 4$  ( $R^2 = 0.88$  on  $n_*^v$  vs  $0.04$  on  $n_*^\alpha$ ). (c) Density-scaling test at  $\rho_0 \in \{1.0, 1.5, 2.22, 3.0\}$  ( $n_*^\alpha = 3$ ). (d) Metric and (e) topological ( $k = 4$ ) heatmaps at  $\rho_0 = 2.22$ , matched scale. (f) Critical threshold  $n_*^{v, \text{crit}}(\rho_0) = -0.73 \rho_0 + 5.81$  ( $R^2 = 0.97$ ; dotted, mean-density prediction  $A \rho_0$ ).

The mean-density prediction is wrong in sign:  $n_{\star}^{v,\text{crit}}$  falls with  $\rho_0$ , the slope  $-0.73$  the mirror image of  $+0.75$ .

The reversed sign most likely traces to the dense-to-dilute density contrast, a phenomenological reading we leave for a first-principles account (App. A). At low  $\rho_0$  the dilute phase is nearly empty, so the dense phase carries a large relative contrast and its local count clears even a demanding  $n_{\star}^v \simeq 5$ ; at high  $\rho_0$  the dilute phase is itself crowded, the contrast weakens, and  $n_{\star}^v$  must drop to stay selective. Extrapolating Eq. (12) to  $\rho_0 \simeq 8$  gives  $n_{\star}^{v,\text{crit}} \rightarrow 0$ : once the two phases are indistinguishable, no threshold buys a coherence gain. The naive alternative is falsified, since if  $n_{\star}^{v,\text{crit}}$  tracked the dense-phase neighbour count  $\langle n_i \rangle_{\text{dense}}$  the two would be equal, yet  $\langle n_i \rangle_{\text{dense}}$  increases with  $\rho_0$  (2.6 at  $\rho_0 = 1.0$  to 5.3 at  $\rho_0 = 3.0$ ), opposite to  $n_{\star}^{v,\text{crit}}$ .

Metric alignment conflates two roles: the sigmoid input  $n_i$  and the partner set are the same object. We pull them apart with the topological variant of Ginelli and Chaté [52], each particle aligning with its  $k$  nearest non-repulsion neighbours while the sigmoid input stays the metric annulus count. Re-running the 25-cell map at  $L = 30$ ,  $\rho_0 = 2.22$  with  $k = 4$  (Fig. 6e), the two-band structure softens: the linear fit on  $n_{\star}^v$  alone drops from  $R^2 = 0.88$  to 0.68 while the  $n_{\star}^{\alpha}$  fit climbs from 0.04 to 0.43, the bivariate fit now splitting its weight nearly evenly, and the sign-flip region retreats, cells once strongly negative under metric alignment turning positive, only the high-high corner staying negative.

So motility-threshold dominance is a property of the metric kernel, not of the two-feedback dynamics. Under metric alignment  $n_{\text{ali}}$  both feeds the sigmoid and selects the alignment partners, so pushing the threshold above typical occupancy switches off the slowdown and the neighbour-driven alignment at once; topological alignment breaks that tie, the  $k = 4$  nearest neighbours handing each particle a baseline alignment the rectification can sharpen even when the motility channel is nearly silent.

The full-mode droplets are larger; are they longer-lived? We track the centroids of the connected dense components over 200 snapshots spaced 50 steps, matching across snapshots by periodic-centroid distance (cutoff 4.0) and particle-count ratio (cutoff 2.0), ten seeds. The mean patch lifetime is 1.60 snapshots in motility-only against 1.57 in the full mode, a difference of  $-0.034$  (95% bootstrap CI  $[-0.10, +0.03]$ , Mann-Whitney  $p = 0.08$ ; not significant); with mean lifetimes barely above the one-snapshot matching resolution, this null bounds any persistence difference rather than resolving sub-snapshot dynamics. The geometric consolidation buys no temporal persistence: the rectification reshapes which cells the dense particles occupy at each instant while leaving cell turnover to the motility dynamics.

Together the four tests pin the rectification down: a spatial consolidation that needs an aggressive motility threshold and a metric kernel to engage, works regardless of the noise-shape threshold, never extends patch lifetimes, and switches off once the dense-to-dilute contrast washes out, piggybacking on the MIPS instability rather than driving the phase separation.

## 4 Discussion

### 4.1 Mechanism and biological reading

What ties the two channels together is the shared sigmoid in  $n_i$ : speed and stability index move in lockstep. Inside a dense patch the sigmoid pushes  $\alpha_i \rightarrow 2$ , replacing the Cauchy bursts that would otherwise scramble a slow particle by Gaussian-like kicks. A rectified kick no longer randomises the local alignment, so the noise-shape channel acts as a fluctuation rectifier *inside* the slow region the motility channel carves out. That is the whole of what it does: it lowers the effective angular noise in the dense phase, and every advantage of the full mode over motility-only is that single effect read out in a different observable. The dense  $g(r)$ , its dynamic counterpart  $C(\tau)$ , the  $s_{\text{sep}}$  surplus and the finite-size susceptibility excess all express it, flat in the dilute quartile, and a matched-noise control removes the full-mode advantage from all of them at once (Table 1): equalising the dense decorrelation  $\Gamma$  drives  $\Delta_9$  from  $+0.45$  to  $+0.02$  and reverses the  $g(r)$  and

$C(\tau)$  gaps as the global retune over-quiets the dilute phase. The two adaptations therefore do not cooperate beyond their noise levels; they share one local route, and at matched noise they simply add. The decoupled-sigmoid tests of Sec. 3.3.4 sharpen the picture: the rectification is a spatial consolidation, not a temporal stabilisation ( $p = 0.08$ ); it needs the metric kernel ( $n_\star^v$  dominance weakening from  $R^2 = 0.88$  to 0.68 under topological alignment); the motility threshold controls it alone; and it switches off once the dense-to-dilute contrast washes out ( $n_\star^{v,\text{crit}} \rightarrow 0$  as  $\rho_0 \rightarrow 8$ ).

The biological reading carries an unambiguous priority. A flock that slows down aggressively reaps the two-feedback cohesion at almost any noise-shape threshold, whereas one with an inert motility channel lets the noise-shape adaptation damage dense-phase coherence. Being a property of the metric kernel (Sec. 3.3.4), this priority applies to short-range metric perceivers such as marching locusts [59, 60] and dense bacterial colonies, where collective motion and cluster formation are set by short-range contacts [61, 62], and not to the topological flocks whose alignment reads a fixed neighbour number and whose correlations are correspondingly scale-free [51, 52, 63].

## 4.2 Literature, limits and outlook

The motility channel inherits directly from Cates and Tailleur [20]: our motility-only ablation reproduces  $\partial \ln v / \partial \ln \rho < -1$  MIPS inside an aligning Couzin flock, with  $s_{\text{sep}}$  rising from  $\simeq 1.2$  for the fixed-speed references to  $\simeq 1.7$ . This is in the spirit of Bi et al. [3]’s density-driven epithelial jamming, but we keep the alignment interaction, so the dense phase is polar rather than amorphous. Within the metric-Vicsek lineage of Vicsek et al. [1], Grégoire and Chaté [43] and Chaté et al. [24] established traveling bands and a weakly first-order transition; the band-soliton index  $b \leq 1.06$  (Fig. S2) shows we never reach that regime, and  $P(\langle \varphi \rangle)$  stays unimodal at  $L = 30$  with  $U_4 \simeq 0.65$  in both modes (this single- $\eta$ , long-trajectory cumulant differs from the  $\eta$ -resolved minimum of Fig. 3d that every mode drives to the same non-diagnostic 0.28–0.31 dip). The Toner–Tu programme [64–66], whose predictions have since been assessed quantitatively against large-scale simulation [67], gives the same continuous polar order but, treating the noise as Gaussian with density-independent variance, does not anticipate the dense-quartile gap of Fig. 4; the  $\Delta a \simeq +0.4$  gap between the full and motility-only slopes is a quantitative target for a density-dependent Toner–Tu closure.

The heavy-tail strand is the second axis. Reynolds [9], Harris et al. [10] and Ariel et al. [11] document Lévy reorientations in foragers, T-cells and swarming bacteria, a strand motivated in part by the search-efficiency argument of Viswanathan et al. [68] and formalised through the anomalous-diffusion framework of Metzler and Klafter [69] and Zaburdaev et al. [12]. Our noise-shape ablation tempers the heavy-tail-as-driver picture: the Cauchy and noise-shape modes carry slopes statistically indistinguishable from zero, so heavy-tailed reorientations on their own grow no size-dependent  $\chi_{\text{max}}(L)$  at our density, the alignment averaging in the dense phase washing the bursts out. The motility channel is necessary, heavy tails alone not sufficient, and what the noise-shape adaptation buys is a lower effective noise inside the dense phase, the one route a matched-noise control isolates. Where Couzin et al. [53] treats the zonal parameters as global constants, we make the stability index a local field on the same kernel; Bechinger et al. [42] and Marchetti et al. [41] review and embed such feedbacks hydrodynamically, but neither identifies the dense-phase Gaussian conversion, because neither couples the noise distribution to the local density, and the one framework general enough to accommodate it leaves that dependence unexplored by design [47].

Each ingredient is documented on its own: a density-dependent speed across active colloids, traffic and jamming [2–4, 20], and environment-dependent reorientation statistics in swimming and swarming bacteria [70–72]. The closest concrete instance of the two responses coexisting under one cue is the dense *E. coli* suspension of Colin et al. [6], where raising cell density both changes the swimming speed and accelerates heading decorrelation through collective reorientations; density also tunes the Lévy exponent of swarming trajectories [7] and the reversal frequency of *M. xanthus* [13]. None of this validates our gate, and it fails to on two counts. The measured density-

dependence of speed in bacterial systems has the opposite sign to ours, rising rather than falling with crowding [6, 7]; and what those experiments resolve is the reorientation *rate*, not the tail shape of the turning distribution, the Gaussian-to-heavy-tailed switch in swarms being controlled by cell aspect ratio rather than by density [7, 73]. The shared sigmoid is therefore a deliberate limiting case, not a parsimony claim about any organism. Its sharpest prediction is a dense-phase narrowing of the turning-angle distribution together with a positive heading-correlation gap at fixed effective decorrelation, which an experiment on a crowding-responsive collective could test, and the sign of the speed response would have to be established alongside it.

The weakest part of the case remains the lever. Extending the scan to  $L = 180$  and  $256$  lowers both motility-active slopes ( $a_{\text{full}} : +1.95 \rightarrow +1.49$ ,  $a_{\text{motility}} : +1.29 \rightarrow +1.04$ ) and shrinks the each-own- $\eta$  excess ( $\Delta_7 = +0.63$  to  $\Delta_9 = +0.45$ ), so the slopes are still drifting on this window rather than settling, and the matched-noise control then shows that even the residual excess is a noise-level effect ( $\Delta_9 \rightarrow +0.02$  on the same nine sizes) rather than a thermodynamic cooperation. The comparison with the standard Vicsek value  $\gamma/\nu = 1.45(2)$  [31, 32] cuts both ways. The full-mode slope agrees with it while the motility-only slope does not, which is reassuring as a consistency check; but with  $\eta^*(L)$  still migrating, no collapse available and the window straddling  $L^* \simeq 128$  [24], we make no claim to have measured a critical exponent. The qualitative split is what survives, and it survives away from  $\rho_0 = 2.22$  as well, so it is not a parameter-window effect either (Table S1). An explicit data collapse still eludes us, and mapping  $g(r)$  at intermediate distances  $r = 2$  to  $5$  would tell whether the rectification carries a characteristic decay length; the absence of a traveling band leaves room for a different cohesive structure, perhaps a slowly-rotating dense droplet, that a real-space analysis at  $L > 30$  should reveal. On the theory side, a Toner–Tu closure with  $v_{\text{eff}}(\rho)$  and a fractional  $D_{\text{rot}}(\rho) \propto \eta^{\alpha(\rho)}$  recovering both the linear  $\Delta g(n_\star^v)$  and the negative slope  $n_\star^{v,\text{crit}}(\rho_0) = -0.73 \rho_0 + 5.81$  would close the mechanistic argument.

## A A heuristic hydrodynamic sketch and the decoupled-sigmoid asymmetry

The model coarse-grains to fields  $\rho(\mathbf{x}, t)$  and  $\mathbf{p}(\mathbf{x}, t) = \rho^{-1} \langle \vec{e}_i \rangle_x$ . The treatment is heuristic, making the rectification analytically transparent rather than deriving it.

### A.1 Continuum fields and conservation

With the continuum neighbour count  $n_i \approx A \rho$ ,  $A \equiv \pi(R_a^2 - R_r^2) \simeq 0.75$ , Eqs. (4)–(6) become density fields,

$$\sigma(\rho) = \frac{1}{1 + \exp[-s(A\rho - n_\star)]}, \quad (13)$$

$$v(\rho) = v_{\text{max}} - \Delta v \sigma(\rho), \quad \Delta v \equiv v_{\text{max}} - v_{\text{min}}, \quad (14)$$

and  $\alpha(\rho)$  likewise interpolating from  $\alpha_{\text{min}}$  to  $\alpha_{\text{max}}$ . Density conservation and the polarisation dynamics read

$$\partial_t \rho + \nabla \cdot [\rho v(\rho) \mathbf{p}] = 0, \quad (15)$$

$$\partial_t \mathbf{p} = -\Gamma(\rho, \eta) \mathbf{p} + \nu \nabla^2 \mathbf{p} - \lambda (\mathbf{p} \cdot \nabla) \mathbf{p} + \dots, \quad (16)$$

with  $\nu, \lambda$  the standard Toner–Tu coefficients [64, 66]. The non-trivial input is that the decorrelation rate  $\Gamma$  depends on  $\rho$  through  $\alpha(\rho)$ .

### A.2 Effective rotational decorrelation

A symmetric  $\alpha$ -stable kick gives  $\langle \cos \xi \rangle = \exp(-\eta^\alpha)$ , so for small  $\eta$

$$\Gamma(\rho, \eta) = 1 - \langle \cos \xi \rangle \simeq \eta^{\alpha(\rho)}. \quad (17)$$

This is  $\eta$  in the dilute (Cauchy) limit and  $\eta^2$  in the dense (Gaussian) one, the noise-shape feedback turning the  $\eta$ -linear Cauchy decorrelation into the much weaker  $\eta^2$  Gaussian one inside the dense phase, the analytic counterpart of the rectification.

Because the kick enters the heading only modulo  $2\pi$ , the form  $\langle \cos \xi \rangle = \exp(-\eta^\alpha)$  is exact rather than small- $\eta$ : it is the mean resultant length of the wrapped symmetric  $\alpha$ -stable law [48–50], so it already integrates the wrapping, and  $\Gamma \rightarrow 1$  as  $\eta^\alpha$  grows, the wrapped kick saturating to the uniform circle. The heavy tail acts through the fraction of kicks that overshoot a half-turn and wrap,  $P(|\xi| > \pi) = 1 - (2/\pi) \arctan(\pi/\eta) \simeq (2/\pi^2) \eta \approx 0.20 \eta$  in the Cauchy limit (1.0%, 2.0%, 4.0%, 6.1% at  $\eta = 0.05, 0.1, 0.2, 0.3$ ), against  $< 10^{-11}$  for the Gaussian one at the same  $\eta$ : a few per cent of every Cauchy step is a near-uniform reorientation the Gaussian channel never produces. Driving  $\alpha$  from 1 to 2 inside the motility-built dense phase removes precisely this wrapped-uniform fraction, the microscopic content of the rectification and the reason the dense-phase turning tail (Fig. 5c) collapses onto a Gaussian core while the dilute gas keeps its Lévy reorientations.

### A.3 MIPS instability and its rectification correction

A homogeneous polar state goes unstable under a longitudinal density perturbation by the Cates–Tailleur mechanism [20] when

$$\left. \frac{\partial \ln v(\rho)}{\partial \ln \rho} \right|_{\rho_0} < -1, \quad (18)$$

which the sigmoidal  $v(\rho)$  meets in a window of width  $1/s$  centred on  $A\rho_0 \simeq n_\star$ . Since  $v$  and  $\alpha$  share the sigmoid, that window coincides with the density at which the rectification fires, so the nucleating dense phase is born already rectified.

The sketch has a real limit. A sequential timing argument predicts only the diagonal ordering of the decoupled map, whereas the data run along horizontal lines of constant  $n_\star^v$  (Fig. 6,  $R^2 = 0.88$  against 0.28 for the timing-offset fit). Neither the horizontal-band structure nor the negative slope  $n_\star^{v,\text{crit}}(\rho_0) = -0.73 \rho_0 + 5.81$  follows from the linear-stability picture; recovering both would need the dense-phase density tied to the MIPS contrast rather than the mean, which we leave to a quantitative theory.

## Code and data availability

The simulation code, all run drivers, the figure-generation scripts and the precompiled data are openly available at <https://github.com/yayayou47/two-feedback-vicsek> under the MIT license. The commit corresponding to this manuscript carries the tag `jstat-submission`, so the exact version underlying every figure can be retrieved unambiguously. All runs used Python 3.12.3, NumPy 2.4.4, SciPy 1.17.1, Numba 0.65.1 (LLVM 0.47.0) and Matplotlib 3.10.9 on Linux x86\_64. The interaction loop is JIT-compiled with Numba’s parallel reduction; per-particle random streams are derived from a single NumPy `Generator` with a per-run seed, so each `(seed, mode, L)` tuple is reproducible bit-for-bit.

## Use of AI-assisted technology

DeepL (DeepL SE) and ChatGPT (OpenAI) were used for language editing, to improve the wording and readability of the manuscript. The authors reviewed and edited all output and take full responsibility for the content of the published article. No text was generated without author verification, and neither tool was used to design the study, to write or modify the simulation and analysis code, or to produce, analyse or interpret any data.



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